

The blueprint for building the Project Aegis prototype:

1. Secure Data Ingestion and PII Sanitization

The hackathon rules explicitly prohibit hosted LLM calls on raw Personally Identifiable Information (PII).

- **The SOTA Technique:** Zero-Shot Generalist Entity Recognition.
- **The Technology:** While your document rightly identifies **Microsoft Presidio** for the pipeline, rely on **GLiNER (Generalist Model for Named Entity Recognition)** for the backend recognizer instead of standard regex or basic spaCy models. GLiNER is current SOTA for zero-shot NER and will adapt seamlessly to localized Indian address formats, PAN formats, and idiosyncratic financial identifiers without needing brittle, hard-coded rules.

2. Agentic Tender Deconstruction (Understanding the Rules)

Standard Retrieval-Augmented Generation (RAG) chunks text, which destroys the structural integrity of complex legal tables and General Staff Qualitative Requirements (GSQR).

- **The SOTA Technique:** Vision-Retriever Document Embedding (Visual RAG).
- **The Technology:** Use **ColPali**. Instead of extracting text, ColPali creates multi-vector representations of the document *images* directly. It preserves the exact visual layout of the General Financial Rules (GFR) clauses and tabular specifications.
- **Vector Database:** Store these visual embeddings in **Qdrant**, which handles high-dimensional vector search beautifully and can be easily spun up locally.

3. OCR-Free Bidder Parsing & Visual Grounding (The Core Engine)

You need to extract structured data from highly unstructured, heterogeneous bidder documents (scanned CA certificates, photos of MSME docs).

- **The SOTA Technique:** Sparse-activated Vision-Language Modeling with Constrained Decoding.
- **The Technology:** **Qwen2.5-VL-7B-Instruct** is the absolute SOTA for native document visual grounding. It inherently understands document layouts and can output exact bounding box coordinates [xmin, ymin, xmax, ymax] for the data it finds.
- **Hardware & Serving:** To maintain data sovereignty and run locally, a 7B parameter model like Qwen2.5-VL fits perfectly within a 20 GB VRAM envelope when served via **vLLM** (which maximizes inference speed).
- **JSON Enforcement:** To prevent the VLM from hallucinating and to ensure it outputs the strict data required for the evaluation matrix, wrap your inference calls using **Outlines** or **Instructor**. This forces the VLM to strictly adhere to a Pydantic schema (e.g., returning only Eligible, Not Eligible, or Need Manual Review alongside the bounding box coordinates).

4. Logic & Orchestration

You need an overarching system to manage the flow between reading the tender, parsing the bidder, and checking the rules.

- **The SOTA Technique:** Graph-based Multi-Agent State Machines.
- **The Technology:** Use **LangGraph**. It allows you to build the Evaluation Agent as a cyclic graph. If the agent extracts a financial figure but the confidence score drops below your deterministic 85% threshold, LangGraph

can automatically route that specific document state to the Human-in-the-Loop (HITL) queue. This guarantees adherence to the non-negotiable rule against silent disqualification.

5. HITL Interface & Deployment

The system must be auditable, explainable at the criterion level, and provide a clear dashboard for the procurement officer.

- **The Technology:** Build a robust, asynchronous backend using **FastAPI** to handle the heavy document processing and LangGraph orchestration.
- **Frontend:** A **React** or **Next.js** frontend is ideal for rendering the PDFs. You can use standard canvas layers over the document image to draw the highlighted bounding boxes using the exact coordinates passed down from Qwen2.5-VL.
- **Deployment:** Wrap the entire architecture (FastAPI backend, Qdrant DB, and frontend) into unified **Docker** containers. This ensures your prototype is a clean, reproducible sandbox that can run offline on deterministically scrambled data, perfectly aligning with the Phase 2 requirements.

This stack bypasses the limitations of legacy systems and provides the verifiable, visually grounded audit trail the CRPF requires.